

An Adaptive Funnel-Based Trading Rule with Transaction-Cost-Safe Exits

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Abstract

This paper formulates a discrete-time, long-only trading rule based on a drift-adjusted predictive funnel for log prices. The strategy enters and exits positions using standardized price deviations, local momentum, adaptive drift and volatility estimates, and an explicit transaction-cost profit threshold. The main formal property is a no-realized-loss guarantee for completed voluntary exits: whenever the algorithm sells under its own exit rule, the executable sale price is high enough to cover proportional buy and sell costs. This guarantee is deliberately narrow. It does not remove mark-to-market drawdown, opportunity cost, indefinite holding risk, model error, taxes, exact exchange microstructure, or finite-horizon liquidation risk. The implementation also includes practical modifications used in the simulation: bounded diagnostic funnels, trailing-profit exits, trend-continuation re-entry, spread/slippage-aware execution approximations, a live sell-limit floor for fixed-fee break-even protection, walk-forward parameter selection, online parameter smoothing, and real market bar data with horizon-dependent sampling intervals.

1 Motivation

The project studies a rule-based version of a common trading idea: hold an asset until price has moved favorably, avoid selling at a realized loss, and re-enter only when the market appears statistically attractive or trend participation is justified. The objective is not to prove global optimality or profitability. The objective is to define a mathematically consistent algorithm, identify exactly what is guaranteed, and evaluate the remaining risks using simulation and real market data.

The strategy is built from four ingredients:

1. a predictive funnel for log prices;
2. a momentum filter for timing entries and exits;
3. a transaction-cost-adjusted profit threshold for voluntary sells;
4. adaptive estimation and walk-forward testing to reduce lookahead bias.

The central distinction is between realized wealth and mark-to-market wealth. The strategy can enforce a no-realized-loss condition on completed voluntary exits, but it cannot prevent the open position from losing value before an exit occurs.

2 Market Model and Predictive Funnel

Let $S_t > 0$ be the asset price observed at discrete times $t = 0, 1, \dots, T$, and let

$$X_t = \log S_t$$

be the log price. The baseline simulation model is

$$X_{t+1} = X_t + \mu + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1),$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$. In the implemented strategy, the trader does not need to know the true μ and σ . Instead, the funnel uses one-step-ahead estimates $\hat{\mu}_{s|s-1}$ and $\hat{\sigma}_{s|s-1}^2$.

After a reference time τ , define

$$C_\tau(t) = X_\tau + \sum_{s=\tau+1}^t \hat{\mu}_{s|s-1}, \quad V_\tau(t) = \sum_{s=\tau+1}^t \hat{\sigma}_{s|s-1}^2, \quad t > \tau.$$

The standardized funnel deviation is

$$Z_{\tau,t} = \frac{X_t - C_\tau(t)}{\sqrt{V_\tau(t)}}.$$

For $k > 0$, the upper and lower log-funnel boundaries are

$$U_\tau(t) = C_\tau(t) + k\sqrt{V_\tau(t)}, \quad L_\tau(t) = C_\tau(t) - k\sqrt{V_\tau(t)}.$$

Thus $Z_{\tau,t} > k$ is equivalent to $X_t > U_\tau(t)$, and $Z_{\tau,t} < -k$ is equivalent to $X_t < L_\tau(t)$.

Remark 1. *At a fixed time under the correctly specified Gaussian baseline, $Z_{\tau,t}$ has a standard normal interpretation. The trading rule, however, monitors the statistic sequentially. Hence k is best viewed as a boundary-width tuning parameter, not as a simple fixed-time significance level.*

3 Momentum and Transaction Costs

For an integer window $\delta \geq 1$, define the finite-difference momentum statistic

$$M_t = X_t - X_{t-\delta}.$$

Positive M_t indicates recent upward movement, while $M_t \leq 0$ indicates that recent upward momentum has stalled or reversed. More precisely, $M_t \leq 0$ means that the log price has not increased relative to $t - \delta$; it does not imply monotone decline during every intermediate step. A generalized momentum version is discussed in the Extensions section.

Suppose buying incurs proportional cost $c_{\text{buy}} \geq 0$ and selling incurs proportional cost $c_{\text{sell}} \in [0, 1)$. A trade entered at τ_b and exited at t has positive after-cost realized multiplier exactly when

$$\frac{(1 - c_{\text{sell}})S_t}{(1 + c_{\text{buy}})S_{\tau_b}} > 1.$$

Equivalently,

$$X_t - X_{\tau_b} > h, \quad h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

The quantity h is the minimum log return required to break even after costs.

4 Signal Price and Executable Price

The funnel, drift estimate, volatility estimate, and momentum statistic are built from the observed bar price. In simulation this is the simulated price; for Yahoo data it is the bar close or adjusted close. This price is denoted

$$S_t^{\text{sig}}, \quad X_t = \log S_t^{\text{sig}},$$

and should be interpreted as a signal price rather than a guaranteed transaction price.

In a live market, buying is executed near the ask and selling is executed near the bid. Since the Yahoo historical bars used in the app do not provide a complete historical order book, the implementation adds a conservative execution layer. With total spread proxy $s \geq 0$ and per-side slippage proxy $\ell \geq 0$, the executable prices are approximated by

$$S_t^{\text{buy}} = S_t^{\text{sig}} \left(1 + \frac{s}{2} + \ell\right), \quad S_t^{\text{sell}} = S_t^{\text{sig}} \left(1 - \frac{s}{2} - \ell\right).$$

Thus the algorithm may learn and generate signals from X_t , but entries are charged at S_t^{buy} , exits are liquidated at S_t^{sell} , and mark-to-market wealth is also reported using the sell-side executable price. In the GUI, s and ℓ are controlled by the spread and slippage sliders in basis points. Setting both to zero recovers the earlier close-price backtest.

Under this execution convention, the profit-buffer condition is evaluated as

$$\log S_t^{\text{sell}} - \log S_{\tau_b}^{\text{buy}} > h.$$

This is stricter than comparing close prices alone: a trade must overcome the modeled bid-ask spread, slippage, and the explicit proportional transaction costs before a voluntary sale is allowed.

4.1 Live Limit-Order Safeguard

The IBKR paper-trading bridge adds an execution-level safeguard for live sell orders. Let a filled buy anchor have execution price $S_{\tau_b}^{\text{fill}}$, let q be the number of shares submitted for sale, let F_{buy} and F_{sell} be fixed dollar commissions assigned to the cycle, and let $m \geq 0$ be a small required net-profit buffer. Before submitting a sell limit order, the bridge computes

$$P_{\min} = \frac{q S_{\tau_b}^{\text{fill}} (1 + c_{\text{buy}}) + F_{\text{buy}} + F_{\text{sell}} + m}{q(1 - c_{\text{sell}})}.$$

The submitted sell limit is then rounded upward to the nearest cent and constrained to satisfy

$$P_{\text{limit}}^{\text{sell}} \geq P_{\min}.$$

Consequently, a sell signal may produce a limit order above the current bid. In that case, the order will not fill immediately; it waits until the market can execute at a price that clears the anchored buy price, fixed fees, proportional costs, and the required profit buffer. In the code, the default fixed fees are $F_{\text{buy}} = 1.00$, $F_{\text{sell}} = 1.03$, and the default live buffer is $m = 0.01$ dollar. This safeguard separates three events: the strategy signal, the order submission, and the broker-confirmed fill. The realized anchor and the buy/sell plot markers are updated only after the broker reports a fill.

5 Trading Rule

The strategy is long-only. In the visualization, the simulation starts with an initial position at the first tradable observation so that the exit logic can be seen immediately. After a sale, the rule alternates between cash and long exposure.

5.1 Voluntary Exit

Let τ_b be the current entry time. The mathematical upper-funnel exit rule is

$$\tau_s = \inf \left\{ t > \tau_b + \delta : Z_{\tau_b, t} > k, \quad M_t \leq 0, \quad \log S_t^{\text{sell}} - \log S_{\tau_b}^{\text{buy}} > h \right\}.$$

The first condition says price is high relative to the predictive funnel; the second says recent upward momentum has weakened; the third ensures the trade is profitable after modeled execution costs and transaction costs.

The Extensions section describes an optional generalized momentum variant that modifies only the second condition while leaving the funnel and profit-buffer conditions unchanged.

5.2 Bounded Diagnostic Funnel

In long trends, the cumulative variance $V_{\tau_b}(t)$ can grow so large that the funnel becomes too wide to trigger practical decisions. The implementation therefore uses a bounded diagnostic reference for signal generation. Given a maximum lookback L , define

$$\rho_t = \max\{\tau_b, t - L\}.$$

The diagnostic statistic used for the funnel test is

$$Z_t^{(L)} = \frac{X_t - C_{\rho_t}(t)}{\sqrt{V_{\rho_t}(t)}}.$$

The original entry price X_{τ_b} is still used for profit accounting. In the current implementation, the default is $L = 60$ bars.

5.3 Trailing-Profit Exit

The implementation also allows a trailing-profit exit. Let

$$H_t = \max_{\tau_b \leq u \leq t} X_u$$

be the highest log price since entry. For a trailing threshold $a > 0$, a trailing exit is allowed when

$$H_t - X_t \geq a \quad \text{and} \quad \log S_t^{\text{sell}} - \log S_{\tau_b}^{\text{buy}} > h.$$

The default implementation uses $a = 0.04$ in log-price units. Since the profit condition is retained, this practical exit does not weaken the no-realized-loss theorem.

5.4 Re-Entry

After a sale at τ_s , the strategy holds cash and uses τ_s as the new reference point. The original symmetric re-entry rule buys after a lower-funnel recovery:

$$Z_{\tau_s, t} < -k, \quad M_t \geq 0.$$

This is a bargain-entry rule. It works naturally in mean-reverting markets but can remain in cash during persistent upward trends. The implementation therefore adds a trend-continuation entry. For $z_{\text{trend}} \geq 0$, buy when either

$$\tau_b^{\text{next}} = \inf \{ t > \tau_s + \delta : (Z_{\tau_s, t} < -k, M_t \geq 0) \quad \text{or} \quad (Z_{\tau_s, t} > z_{\text{trend}}, M_t > 0) \}.$$

The default value is $z_{\text{trend}} = 0.35$, intentionally smaller than k . The first condition seeks a recovered discount; the second permits participation in an ongoing upward trend. At the current Layer 2 stage, z_{trend} is treated as a fixed design constant; in later calibration work it should either be selected by walk-forward validation or made time-varying with the entry/exit thresholds.

6 No-Realized-Loss Property

Theorem 1 (Transaction-cost-safe voluntary exit). *Suppose a position is entered at τ_b and voluntarily exited at a finite time τ_s by either the funnel exit or the trailing-profit exit. If the exit rule includes the condition*

$$\log S_{\tau_s}^{\text{sell}} - \log S_{\tau_b}^{\text{buy}} > h,$$

then the after-cost realized multiplier satisfies

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}^{\text{sell}}}{(1 + c_{\text{buy}})S_{\tau_b}^{\text{buy}}} > 1.$$

Proof. By definition of h ,

$$\log S_{\tau_s}^{\text{sell}} - \log S_{\tau_b}^{\text{buy}} > \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

Exponentiating gives

$$\frac{S_{\tau_s}^{\text{sell}}}{S_{\tau_b}^{\text{buy}}} > \frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}}.$$

Multiplying by $(1 - c_{\text{sell}})/(1 + c_{\text{buy}})$ proves the claim. $\square$

Remark 2. *The theorem only applies to completed voluntary exits. It does not apply to forced liquidation at a finite horizon, margin liquidation, inability to execute, taxes, slippage beyond the modeled execution proxy, or a position that remains open indefinitely.*

7 Wealth and Performance Diagnostics

Let W_t^r denote realized wealth, updated only after completed exits. While holding a position entered at τ_b , the implementation reports after-cost mark-to-market wealth as

$$W_t^m = W_{\tau_b}^r \frac{(1 - c_{\text{sell}})S_t^{\text{sell}}}{(1 + c_{\text{buy}})S_{\tau_b}^{\text{buy}}}.$$

The buy-and-hold benchmark is computed from the same initial exposure convention:

$$W_t^{\text{bh}} = W_0 \frac{(1 - c_{\text{sell}})S_t^{\text{sell}}}{(1 + c_{\text{buy}})S_{\tau_0}^{\text{buy}}}.$$

Let

$$N_{\text{trades}} = \#\{i : \tau_{s,i} \leq T\}$$

be the number of completed trades by horizon T . For completed trades, define the average holding time

$$\bar{\tau}_{\text{hold}} = \frac{1}{N_{\text{trades}}} \sum_{i=1}^{N_{\text{trades}}} (\tau_{s,i} - \tau_{b,i}),$$

when $N_{\text{trades}} > 0$.

The dashboard reports realized return, mark-to-market return, buy-and-hold return, benchmark gap,

$$G_T = \frac{W_T^m}{W_T^{\text{bh}}} - 1,$$

maximum drawdown,

$$D_{\min} = \min_{0 \leq t \leq T} \left(\frac{W_t}{\max_{0 \leq s \leq t} W_s} - 1 \right),$$

exposure, N_{trades} , $\bar{r}_{\text{hold}}$, Sharpe ratio, and Calmar ratio. For bar frequency B periods per year, the annualized Sharpe ratio is

$$\text{Sharpe} = \sqrt{B} \frac{\bar{r}}{s_r}, \quad r_t = \frac{W_t - W_{t-1}}{W_{t-1}},$$

and the Calmar ratio is

$$\text{Calmar} = \frac{R_{\text{ann}}}{|D_{\min}|}.$$

The implementation uses $B = 252$ for daily bars, $B = 252 \cdot 6.5$ for hourly bars, and $B = 252 \cdot 26$ for 15-minute bars. These annualization constants assume U.S. equity trading hours; they should be changed for assets or venues with different trading calendars.

8 Adaptive Estimation

The trading rule uses estimates available before observing the next return, preventing lookahead in the displayed funnel.

8.1 Drift

Let $r_t = X_t - X_{t-1}$. The drift is modeled as a slowly moving latent state,

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q),$$

with observation equation

$$r_t = \mu_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \sigma^2).$$

The Kalman one-step prediction is

$$\hat{\mu}_{s|s-1} = \hat{\mu}_{s-1|s-1}, \quad P_{s|s-1} = P_{s-1|s-1} + q,$$

and the update after observing r_s is

$$K_s = \frac{P_{s|s-1}}{P_{s|s-1} + \sigma^2},$$

$$\hat{\mu}_{s|s} = \hat{\mu}_{s|s-1} + K_s(r_s - \hat{\mu}_{s|s-1}), \quad P_{s|s} = (1 - K_s)P_{s|s-1}.$$

The UI drift-adaptation control changes q : larger values make the drift estimate react faster to recent returns.

8.2 Volatility

The conditional variance forecast uses a GARCH-style recursion. With innovation

$$e_s = r_s - \hat{\mu}_{s|s-1},$$

define

$$\hat{\sigma}_{s|s-1}^2 = h_s, \quad h_{s+1} = \omega + \alpha e_s^2 + \beta h_s,$$

where $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$. The implementation chooses

$$\omega = (1 - \alpha - \beta)\bar{\sigma}^2$$

from an initial variance estimate $\bar{\sigma}^2$. The funnel variance accumulates the one-step forecasts,

$$V_\tau(t) = \sum_{s=\tau+1}^t \hat{\sigma}_{s|s-1}^2.$$

The Extensions section describes an optional regime-filtering layer based on a three-state hidden Markov model. In that mode, the same trading rule is retained but the one-step funnel inputs are filtered through latent regime probabilities.

9 Backtesting and Real-Data Implementation

The offline walk-forward procedure selects parameters using only past data. For fold j , let $\mathcal{T}_j$ be the training window and $\mathcal{O}_j$ the following out-of-sample window. For candidate parameter vector θ , the training objective used in the implementation has the form

$$J_j(\theta) = R_{j,\text{train}}^m(\theta) + D_{j,\text{train}}^m(\theta) + \lambda G_{j,\text{train}}(\theta),$$

where R^m is mark-to-market return, $D^m \leq 0$ is maximum drawdown, and G is the benchmark gap. The selected parameter is

$$\hat{\theta}_j = \arg \max_{\theta \in \Theta} J_j(\theta).$$

It is then evaluated on $\mathcal{O}_j$ without using future information. The current implementation uses $\lambda = 0.15$, so benchmark-relative performance influences selection but does not dominate absolute mark-to-market return or drawdown.

In the accompanying calibration reports, the parameter search has been explored up to four dimensions: the baseline two-dimensional search over (k, δ) , three-dimensional expansions (k, δ, x) , and four-dimensional expansions (k, δ, x, y) for selected design parameters. Moving beyond four dimensions is possible in principle but becomes computationally expensive and less interpretable. The current project therefore treats four-dimensional calibration as a practical upper bound for Layer 2 sensitivity analysis, rather than as a claim that the full parameter vector has been globally optimized.

For real market data, the app uses a continuous online variant. Instead of resetting wealth and position at every fold, it keeps one account alive. Every update window, recent bars generate a suggested parameter vector $\tilde{\theta}_j$, and the live parameter is smoothed:

$$\theta_j = (1 - \rho)\theta_{j-1} + \rho\tilde{\theta}_j, \quad 0 < \rho \leq 1.$$

Integer parameters such as δ are rounded and constrained to the candidate grid. The current implementation uses learning rate $\rho = 0.25$ and update window 5 bars. This gives a more realistic

animation: the account learns from recent data while position state, realized wealth, and mark-to-market wealth continue without reset.

The implementation can fetch Yahoo Finance chart data for several symbols, including SPY, NVDA, TSLA, MSFT, AAPL, AMZN, GOOGL, META, and custom tickers. The selected horizon determines the bar interval:

lookback	bar interval
5 years	1 day
2 years	1 day
1 year	1 day
1 month	1 hour
1 week	15 minutes

This avoids having only a few observations in short-horizon animations. Intraday results should be interpreted cautiously because microstructure noise, bid-ask spreads, execution latency, and Yahoo interval availability matter more at shorter bar lengths.

10 Empirical Illustration

As a basic Layer 2 validation, the current implementation was run on SPY using Yahoo adjusted price data, a 5-year lookback, 1-day bars, zero transaction costs, drift-adaptation variance $q = 10^{-7}$ (the default UI value), $\lambda = 0.15$, $\rho = 0.25$, $L = 60$, $a = 0.04$, and $z_{\text{trend}} = 0.35$. The purpose of this table is not to claim profitability, but to show that the framework produces the diagnostics required to compare the strategy against buy-and-hold. Because the data window is fetched relative to the current date and the UI sliders remain editable, small differences can occur if the run date or parameters differ.

Quantity	SPY result
Bars	1255
Start price	388.18
End price	720.65
Realized return	53.10%
Mark-to-market return	65.93%
Buy-and-hold return	85.65%
Benchmark gap	−10.62%
Strategy max drawdown	−24.64%
Buy-and-hold max drawdown	−24.50%
Exposure	92.60%
N_{trades}	9
$\bar{\tau}_{\text{hold}}$	117.11 bars
Sharpe ratio	0.69
Calmar ratio	0.43
Online parameter updates	250
Final (k, δ)	(0.80, 2)

In this example, the strategy earned a positive mark-to-market return but underperformed buy-and-hold. This is a useful outcome rather than a failure of the framework: it demonstrates why benchmark gap, drawdown, exposure, trade count, Sharpe ratio, and Calmar ratio must be reported together. The no-realized-loss theorem concerns completed voluntary exits only; it does not imply benchmark outperformance.

11 Extensions

The first implemented Layer 3 extensions are generalized momentum, a three-state regime HMM filter, and randomized stopping. All three are available in the GUI for simulated paths and real Yahoo Finance data, so they can be compared against the fixed baseline and the online optimization mode on the same selected price path.

11.1 Generalized Momentum

Generalized momentum replaces the fixed exit threshold $M_t \leq 0$ with the holding-time-dependent threshold

$$f(t; \tau_b) = -\frac{c}{\sqrt{t - \tau_b}}, \quad c \geq 0.$$

The baseline rule is recovered when $c = 0$. For $c > 0$, early exits require a stronger negative momentum signal, while later exits gradually relax back toward the original $M_t \leq 0$ condition. The funnel condition $Z_{\tau_b, t} > k$ and the transaction-cost profit buffer $X_t - X_{\tau_b} > h$ are unchanged, so the no-realized-loss theorem still applies to completed voluntary exits. The parameter c is tunable in the GUI.

11.2 Regime HMM Filtering

The HMM extension is best viewed as a filtering layer for the funnel, not as a separate trading rule. The buy, sell, re-entry, and profit-buffer logic remain the same. What changes is the estimate of the next-step drift and variance used inside $C_\tau(t)$ and $V_\tau(t)$.

Let

$$R_t \in \{1, 2, 3\}$$

denote a latent regime, interpreted in the implementation as calm, mixed, and stress states. Conditional on $R_t = j$, returns satisfy

$$r_t \mid R_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2),$$

and R_t evolves according to a persistent Markov transition matrix. The implementation initializes (μ_j, σ_j^2) from an early training segment by sorting returns into volatility buckets, then updates regime probabilities online by forward filtering.

Let

$$\pi_{t|t-1}(j) = P(R_t = j \mid r_{1:t-1})$$

be the predicted regime probability before observing r_t . The filtered one-step drift input is the mixture mean

$$\hat{\mu}_{t|t-1}^{\text{HMM}} = \sum_{j=1}^3 \pi_{t|t-1}(j) \mu_j,$$

with filtered mixture variance

$$\hat{\sigma}_{t|t-1}^{2, \text{HMM}} = \sum_{j=1}^3 \pi_{t|t-1}(j) (\sigma_j^2 + \mu_j^2) - \left(\hat{\mu}_{t|t-1}^{\text{HMM}} \right)^2.$$

These filtered inputs replace the Kalman/GARCH inputs inside the diagnostic funnel. The GUI also displays the most likely regime and posterior probability during playback, making the extension interpretable as the path evolves.

11.3 Randomized Stopping

Randomized stopping turns the deterministic sell decision into a probabilistic gate. It is applied only after the usual exit eligibility conditions have already been satisfied. Let

$$E_t = \mathbf{1}\{Z_{\tau_b,t} > k, M_t \leq 0, X_t - X_{\tau_b} > h\}$$

denote the baseline funnel-exit eligibility event, with the analogous trailing-profit eligibility event also allowed in the implementation. When an eligible exit occurs, the extension computes a selling probability

$$p_t = \frac{1}{1 + \exp[-(\alpha + \beta A_t)]}, \quad \beta > 0,$$

where $A_t \geq 0$ measures how far the active exit signal is beyond its threshold. For a funnel exit the implementation uses

$$A_t = Z_{\tau_b,t} - k,$$

and for a trailing-profit exit it uses the normalized excess drawdown beyond the trailing threshold. A Bernoulli random variable

$$B_t \sim \text{Bernoulli}(p_t)$$

is then sampled, and the sell is executed only when $B_t = 1$.

This extension does not remove the transaction-cost profit buffer. Therefore, any realized voluntary sell that is actually executed still satisfies $X_t - X_{\tau_b} > h$, so the no-realized-loss theorem remains valid for completed voluntary exits. The randomization only changes the timing of eligible exits; it can delay a sale and therefore can increase or decrease mark-to-market risk depending on the path. In the current implementation, $\alpha = 0$, $\beta = 4$, and a fixed random seed is used for reproducible GUI comparisons on the same path.

Further Layer 3 work may apply the same time-dependence idea to z_{trend} , making trend re-entry stricter or looser depending on time since the last sale. These additions introduce new calibration degrees of freedom and should be tested by walk-forward validation before being promoted to the main rule.

12 Limitations

The strategy is not risk-free. The main limitations are:

- **Mark-to-market risk:** open positions can lose value before an exit occurs.
- **Holding-time risk:** the algorithm may hold a losing or stagnant position for a long time.
- **Forced-exit risk:** liquidation outside the voluntary exit rule can realize losses.
- **Opportunity cost:** the strategy may underperform buy-and-hold, especially in persistent trends or if re-entry is too conservative.
- **Estimation error:** drift is difficult to estimate, and volatility models are misspecified in real markets.
- **Data and execution risk:** public chart data may contain gaps or limits. The implementation includes only a simple spread/slippage proxy, not full order-book dynamics, taxes, market impact, latency, or guaranteed execution.
- **Parameter sensitivity:** k , δ , q , L , a , and z_{trend} affect both returns and risk.

13 Conclusion

The Layer 1–2 strategy now consists of a formal long-only trading rule, an implementation-aligned no-realized-loss theorem for voluntary exits, adaptive drift and volatility estimation, practical bounded and trailing exit logic, two-mode re-entry, walk-forward evaluation, and real-data testing across multiple lookbacks and bar intervals. The guarantee is intentionally limited but useful: completed voluntary exits are transaction-cost-safe by construction. The remaining question is empirical quality, which must be judged through benchmark-relative return, mark-to-market draw-down, exposure, trade frequency, Sharpe ratio, Calmar ratio, and robustness across assets and sampling frequencies.